PHY4605 Computational Methods in Physics
Learning Note | Week 6

Numerical Differentiation: Estimate a physical rate and test the step size

Estimate a physical rate from nearby values, then check how well the numerical derivative matches a known reference.

Physical question. A projectile follows a familiar vertical-position model. How can two nearby position values estimate its instantaneous vertical velocity at $t = 1.0\,\mathrm{s}$?

Learning outcomes

1. interpret a derivative as a physical rate and preserve its unit;
2. trace a forward-difference estimate for several supplied step sizes; and
3. validate a numerical derivative against a calculus-based reference and simple error.

How to use this note

Read the physical rate and its unit before the MATLAB code. Pause at each prediction and validation checkpoint before running the lecture demonstration. Keep the model, sample locations, step size, numerical estimate, and reference comparison connected.

1. A derivative is a physical rate

In physics, a derivative is not only a calculus symbol. It describes how one physical quantity changes when another changes. If vertical position y is measured in metres and time t in seconds, then

$$\frac{dy}{dt}$$

has unit metres per second. It is a vertical velocity.

The same idea appears elsewhere: dT/dt is a heating or cooling rate, and dV/dx is a spatial gradient of electric potential. Week 6 focuses on estimating such rates numerically when we have model values rather than directly using a symbolic derivative.

Prediction checkpoint. At $t = 1.0\,\mathrm{s}$, a projectile launched upward at $20\,\mathrm{m\,s^{-1}}$ should still be moving upward, but more slowly because gravity acts downward. Therefore dy/dt should be positive and less than $20\,\mathrm{m\,s^{-1}}$.

2. From a tangent slope to two nearby values

A tangent is the straight line that matches a curve at one point; its slope gives the exact rate at that point. A secant is the straight line joining two distinct points; its slope gives the average rate across the interval. A forward difference replaces the tangent slope by this secant slope through two nearby points:

$$\left.\frac{dy}{dt}\right|_{t=t_0} \approx \frac{y(t_0 + h) - y(t_0)}{h}.$$

Here t_0 is the target time and $h > 0$ is the time separation between the two samples.

In words:

change in position ÷ change in time → approximate velocity.

The numerator has unit metres and the denominator has unit seconds, so the derivative estimate has unit m/s.

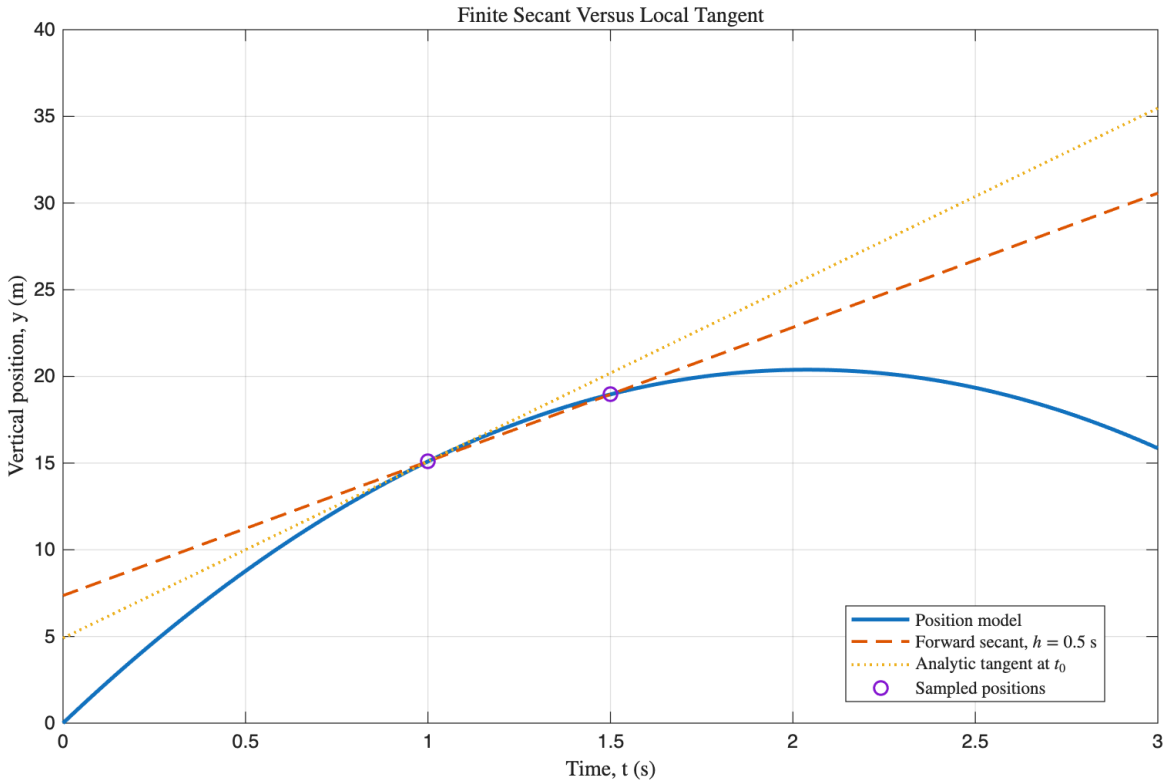


Figure 1: MATLAB-generated geometry for the Week 6 vertical-motion model. The forward-difference secant over a finite step approximates the tangent slope at t_0.

3. Reuse a familiar vertical-motion model

Use

$$y(t) = y_0 + v_0 t - \frac{1}{2} g t^2,$$

with

$$y_0 = 0\,\mathrm{m}, \qquad v_0 = 20\,\mathrm{m\,s^{-1}}, \qquad g = 9.81\,\mathrm{m\,s^{-2}}.$$

We want the derivative at

$$t_0 = 1.0\,\mathrm{s}.$$

The derivative from calculus is

$$v(t) = \frac{dy}{dt} = v_0 - gt,$$

This calculus result provides an analytic reference: a value obtained by differentiating the formula rather than by using the numerical difference. It is an independent check, not the numerical method itself. At the target time,

$$v(1.0) = 20 - 9.81(1.0) = 10.19\,\mathrm{m\,s^{-1}}.$$

4. One forward difference by hand

For $h = 0.5\,\mathrm{s}$,

$$y(1.0) = 15.095\,\mathrm{m}, \qquad y(1.5) = 18.96375\,\mathrm{m}.$$

Therefore

$$v_{\mathrm{FD}}(1.0) = \frac{18.96375 - 15.095}{0.5} = 7.7375\,\mathrm{m\,s^{-1}}.$$

The sign and unit are physically sensible, but the estimate is lower than the analytic value $10.19\,\mathrm{m\,s^{-1}}$. The finite interval from 1.0 to 1.5 s averages a changing velocity rather than sampling an exact tangent.

5. Pseudocode before MATLAB

1. choose the target time t_0 and one supplied positive step h;
2. evaluate the position model at t_0;
3. evaluate the same model at $t_0 + h$;
4. subtract the two position values;
5. divide by h to obtain the forward-difference estimate;
6. compare with the analytic derivative at the same t_0; and
7. repeat for the next supplied step size.

6. Starter MATLAB calculation

Listing 1: Starter MATLAB calculation of forward differences for supplied time steps

```
y0_m = 0;
v0_mps = 20;
g_mps2 = 9.81;
t0_s = 1.0;
h_s = [0.5 0.2 0.1 0.05];
y_now_m = y0_m + v0_mps*t0_s - 0.5*g_mps2*t0_s^2;
y_forward_m = y0_m + v0_mps*(t0_s+h_s) ...
    - 0.5*g_mps2*(t0_s+h_s).^2;
v_forward_mps = (y_forward_m-y_now_m)./h_s;
```

The important line is the final one: the change in position is divided element-by-element by the corresponding time step. The target time and physical model remain fixed while only h changes.

7. Compare a small set of step sizes

The supplied step sizes give:

h (s)	v_{FD} (m/s)	absolute error (m/s)	relative error (%)
0.50	7.7375	2.4525	24.0677
0.20	9.2090	0.9810	9.6271
0.10	9.6995	0.4905	4.8135
0.05	9.94475	0.24525	2.4068

For this smooth model and this supplied range, the estimate approaches the analytic value as h becomes smaller. The lecture Live Script also supplies an error-versus-step-size graph as a guided extension. Read the supplied graph as evidence from one model and finite range, not as proof that arbitrarily tiny steps are always better.

8. Required validation: compare with the analytic derivative

The required validation is a direct comparison at the same target time:

$$v_{\text{exact}}(1.0) = 10.19\,\text{m}\,\text{s}^{-1}.$$

For $h = 0.05\,\text{s}$,

$$v_{\text{FD}} = 9.94475\,\text{m}\,\text{s}^{-1},$$

so the absolute error is

$$|9.94475 - 10.19| = 0.24525\,\text{m}\,\text{s}^{-1},$$

and the relative error is approximately

$$\frac{0.24525}{10.19} \times 100\% \approx 2.41\%.$$

Validation statement. The smallest supplied forward step gives a positive velocity with the correct unit and is within about 2.41% of the known analytic velocity. This supports the numerical estimate for the stated model and target time.

9. Diagnose a runnable but wrong calculation

A common defect is

Listing 2: Example of an incorrect derivative denominator

```
wrong_mps = (y_next_m - y_now_m)/t0_s;
```

The code runs, but the denominator is wrong. The two sampled positions are separated by h, not by t_0. Numerical correctness requires the denominator to represent the actual input interval between the two model values.

Responsible-AI caution. AI-generated differentiation code can be syntactically valid while using the wrong interval, mixing units, differentiating the wrong quantity, or comparing values at different target points. Check the two sample locations, denominator, derivative unit, and reference point before accepting the result.

Guided extension: central difference

A supplied central-difference calculation uses values on both sides of the target point:

$$\left.\frac{dy}{dt}\right|_{t=t_0} \approx \frac{y(t_0 + h) - y(t_0 - h)}{2h}.$$

For this quadratic position model, the supplied central-difference result happens to reproduce the analytic derivative to the precision available in the computer's stored numbers. This special agreement should not be generalised to every physical model. Read the supplied result and interpret the comparison; independent implementation is not required here.

A supplied error-versus-step-size curve may also be inspected qualitatively. A computer stores numbers with finite precision, so subtracting nearly equal values can lose useful digits; extremely small steps are therefore not guaranteed to improve every calculation. Truncation error here means the error caused by replacing the tangent with a finite-step formula. Round-off error is the small numerical change caused by finite-precision storage and arithmetic. Detailed error-order, truncation, and round-off analysis is an optional extension, not a required Week 6 task.

Optional extension

Optional distinction-level extensions include deriving error order, analysing truncation and round-off in detail, or implementing higher-order difference formulas. These activities are not required. Removing this section leaves the complete required physics-to-rate and validation pathway intact.

Three takeaways

1. A numerical derivative estimates a physical rate from nearby values and must carry the correct derivative unit.
2. A forward difference uses a finite secant, so the choice of step size affects the approximation.
3. Validate the estimate against an analytic or supplied reference and report simple error before trusting it.

Preparation for the practical. Be ready to transfer the same reasoning to vertical velocity, electric field from a potential gradient, and a cooling rate. For every context, identify the quantity being differentiated, the input variable, the derivative unit, the two sample locations, the step size, and the validation reference.